

# Gradient descent prunes PING via Dale's-law clamping

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The trained networks this entry uses are produced once in the shared training hub, nb022 (Training), and reused here rather than retrained.

## Abstract

nb025 freezes the recurrent inhibitory weights  $W^{EI}, W^{IE}$  at biophysical values. Unfreeze them and Adam does *not* rediscover PING: from any initialisation — canonical, zero, or 10% of canonical — training prunes most  $W^{EI}$  entries below zero, the Dale’s-law forward-pass clamp turns them into structural zeros, and the network collapses to dense E firing with I silent, at  $\approx 88\%$  accuracy ( $\approx 2$  pp *above* the frozen PING control). PING is a structural prior the freeze *imposes*, not one gradient descent recovers on its own.

## Method

**Architecture.**  $N_E = 1024$  excitatory,  $N_I = 256$  inhibitory, mem-mean readout, Dale’s law enforced. Hyperparameters match the nb025 PING baseline: Adam at  $\text{lr } 4 \times 10^{-4}$ , batch 256, surrogate slope 1,  $W_{\text{in}} \sim \mathcal{N}(1.2, 0.12)$  at 95% sparsity, gradient norm clipped to 1.0,  $\Delta t = 0.1$  ms,  $T = 200$  ms, no firing-rate regulariser.

**Sweep.** Four conditions  $\times$  three seeds (42, 43, 44), medium tier (1600 train / 400 test MNIST, 100 epochs). Only the initial  $(W^{EI}, W^{IE})$  and the trainable-or-not flag vary:

| Condition                    | $W^{EI}, W^{IE}$ init     | Trainable? |
|------------------------------|---------------------------|------------|
| <i>frozen_ping</i> (control) | canonical biophysical     | no         |
| <i>trainable_ping_init</i>   | canonical biophysical     | <b>yes</b> |
| <i>trainable_zero_init</i>   | 0 (COBA-equivalent start) | <b>yes</b> |
| <i>trainable_small_init</i>  | $0.1 \times$ canonical    | <b>yes</b> |

Canonical biophysical means  $W^{EI} \sim \mathcal{N}(1.0, 0.1)$   $\mu\text{S}$ ,  $W^{IE} \sim \mathcal{N}(2.0, 0.2)$   $\mu\text{S}$  at  $N_I = 256$ , fan-in-normalised — so the trainer reports per-edge means of  $\approx 0.0010$  and  $\approx 0.0078$ . “PING is on” means the inhibitory loop is alive: I fires at a healthy rate and paces a gamma rhythm; “the loop is gone” means I is silenced and E fires densely (plain COBA). The firing rates are the cleanest read of which regime a trained network lands in.

## Results

The whole sweep collapses to one picture. In the (E firing rate, I firing rate) plane, a network “found PING” if it sits in the low-E / high-I corner, and “lost the loop” if it sits in the dense-E / silent-I corner.

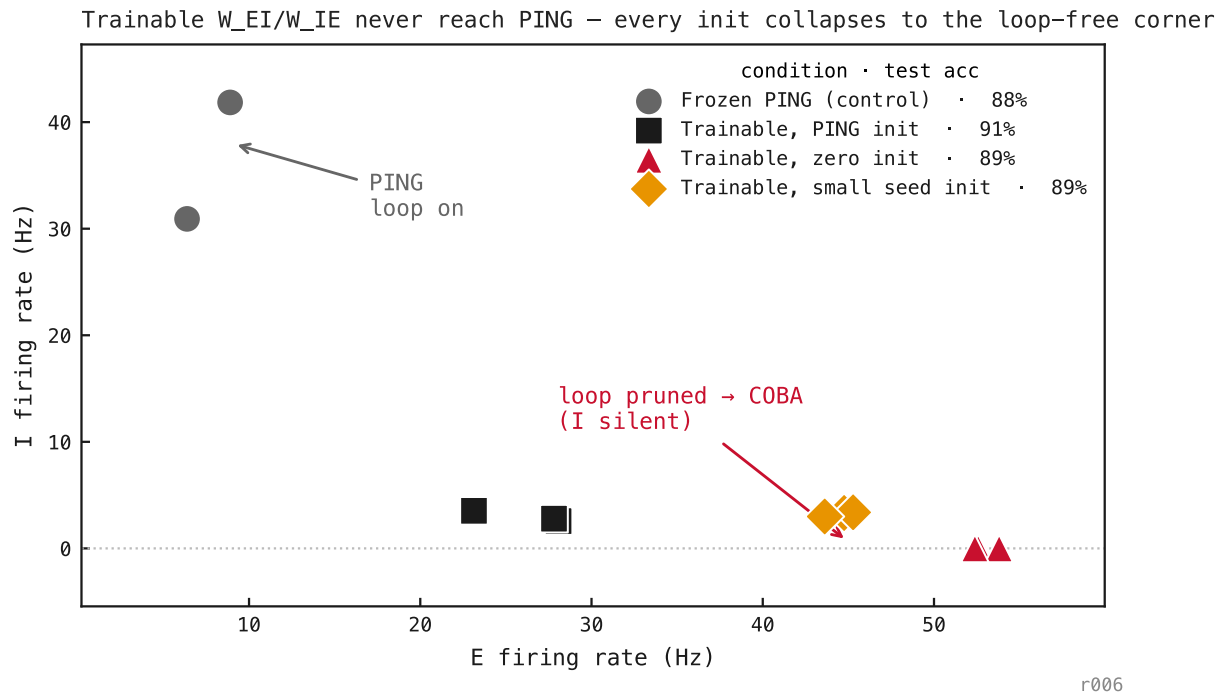


Figure 149: Each point is one trained network ( $4 \text{ conditions} \times 3 \text{ seeds}$ ). The **frozen control** (grey) sits in the PING corner —  $E \approx 8 \text{ Hz}$ ,  $I \approx 38 \text{ Hz}$ , the inhibitory loop alive. **Every trainable condition** — started at canonical PING, at zero, or at  $0.1 \times$  canonical — collapses to the same dense-E / silent-I corner ( $E \approx 45 \text{ Hz}$ ,  $I \approx 0$ ), at  $\approx 88\%$  accuracy,  $\approx 2 \text{ pp}$  *above* the control. Where the loop *starts* makes no difference; only whether it is allowed to train. Gradient descent never reaches PING, and the task slightly prefers it gone.

## Training curves — the loop pruned, epoch by epoch

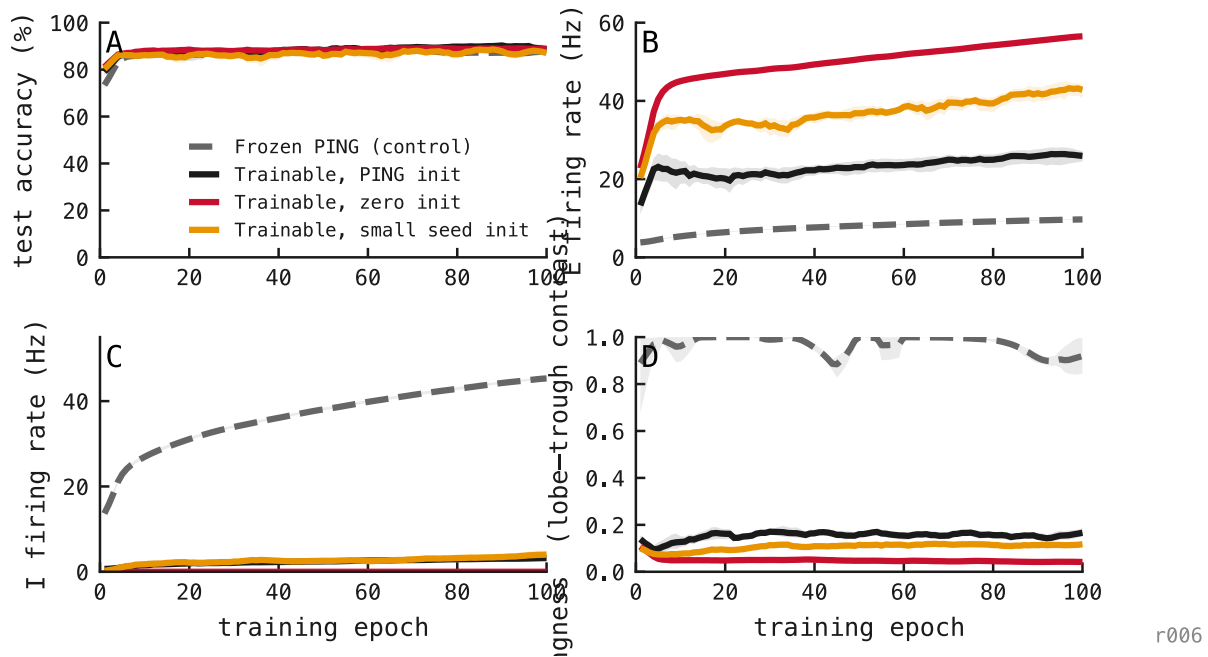


Figure 150: Per-epoch metrics from the runs themselves, coloured by init — PING init (black), zero init (red), small-seed init (amber), frozen-PING control (grey dashed); 5% of MNIST, 100 epochs, 3 seeds each. **Accuracy** — both reach  $\approx 88-90\%$  and track each other; pruning the loop costs nothing (the trainable runs even edge slightly above). **E rate** — the trainable runs climb to  $\approx 25-56$  Hz as the loop releases the excitatory population, while the frozen control stays gated near 8 Hz. **I rate** — every trainable run drops to  $\approx 0$  within a few epochs (the inhibitory loop is pruned), while the frozen control's I rate *rises* to  $\approx 45$  Hz as its readout trains. **Pingness** — the nb054 lobe-trough contrast: the frozen control climbs to  $\approx 0.95$  (the rhythm sharpens) while every trainable init collapses to  $\approx 0.1-0.2$ . The residual floor is the metric's known low-rate inflation under shared input (nb054), not a surviving rhythm — the frozen-vs-trainable gap is the signal. The loop dies early, from every start, with no accuracy penalty.

## Phase portrait — there is only one attractor under training

The four per-epoch panels above tell the story but separate the two state variables that matter. Putting *E rate* and *pingness* on perpendicular axes shows the trajectories of training directly, and the geometry rules out a misreading: it is *not* the case that PING and COBA are two attractors of the same dynamics, with the architecture deciding the basin. Under training there is **one** attractor — the COBA corner — and the PING state only persists because freezing the loop zeroes its gradients.

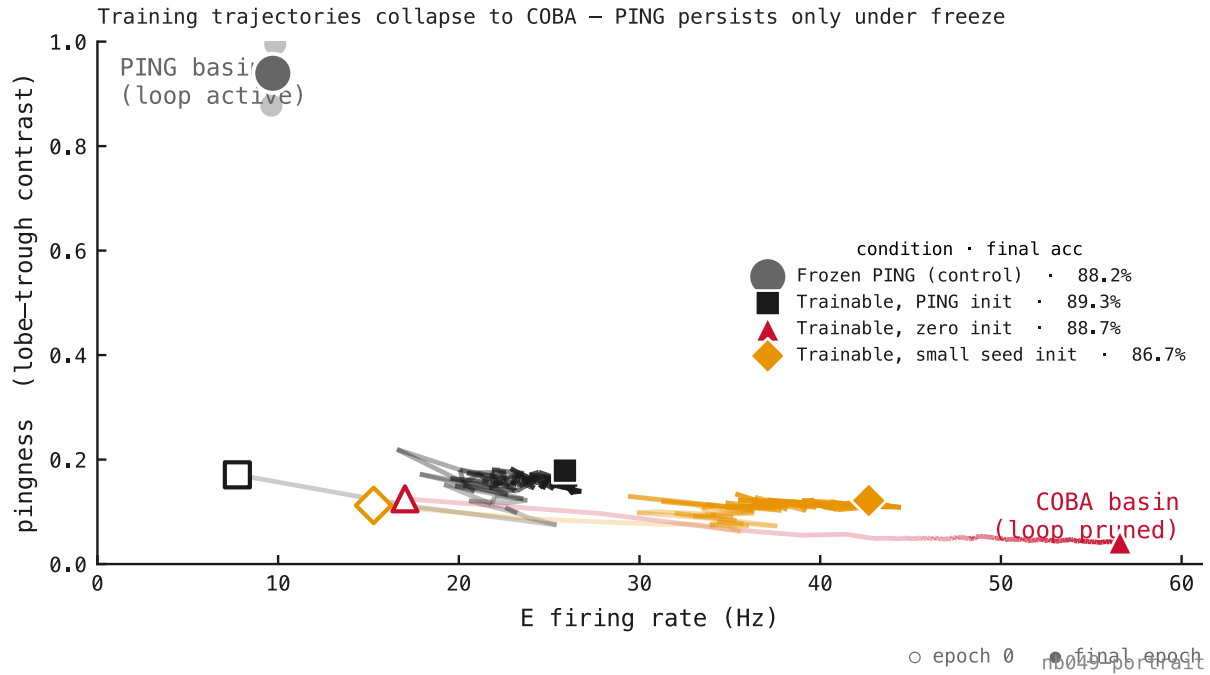


Figure 151: Each trainable trajectory is the per-epoch mean across 3 seeds, alpha-ramped along the epoch axis so the direction of flow reads off the line itself; open markers are epoch 1 (the first logged epoch), filled markers are epoch 100. **Frozen PING (grey)** — sits in the upper PING basin from start to finish at pingness  $\approx 0.86 \rightarrow 0.94$ ; drifts a little in E rate ( $\approx 4 \rightarrow 10$  Hz) as the feedforward and readout weights train, but the loop weights themselves can't move by construction. **Trainable PING init (black)** — was initialised with canonical biophysical loop weights, but by the time the first metric is logged (after epoch 1) the loop has already been pruned: pingness  $\approx 0.17$  from the start. The trajectory then walks rightward along the COBA floor with the others. **Trainable zero init (red)** and **small-seed init (amber)** — start at the floor and stay there, E rate climbing as the feedforward layer learns the task. Every legend entry lands at 87–89% accuracy, so the move costs nothing. The reading is sharper than the time-series panels suggest: the empty band between pingness  $\approx 0.2$  and  $\approx 0.85$  is the *whole story* — gradient descent flies through it within one epoch and the metric never catches it mid-flight. There is no separatrix between PING and COBA under training because there is no slow trajectory through the gap. The freeze isn't preventing slow erosion; it's preventing instant collapse.

### Accuracy–rate trajectory — same destination, different spike economy

Figure 3 puts pingness on its own axis. Putting pingness on the *colour* axis instead and replaying training as (E rate, accuracy) trajectories — the nb025 accuracy–rate frontier given a time axis — gives the third reading of the same data: every condition reaches the same final test accuracy, but along very different routes, and only one of them is doing PING while it gets there.

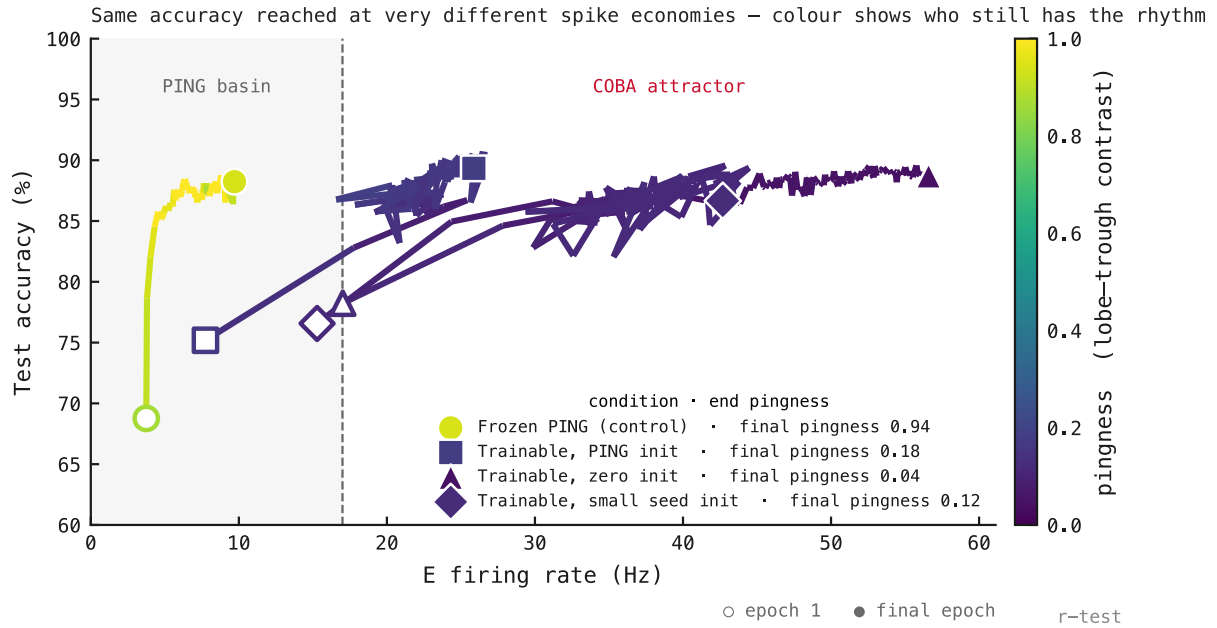


Figure 152: Each trajectory is the per-epoch mean across 3 seeds; per-segment colour is that epoch’s pingness on a viridis 0→1 scale (colourbar at right). Open markers are epoch 1, filled markers are epoch 100. The frontier *with time* tells three things at once. **Same destination** — all four conditions land at  $\approx 87\text{--}89\%$  accuracy by epoch 100, regardless of init or whether the loop trains. **Different routes** — the frozen control climbs accuracy almost vertically with little change in E rate ( $\approx 4 \rightarrow 10$  Hz); the trainable conditions all bend right and up, the biggest one (zero init) reaching the final accuracy at  $E \approx 57$  Hz,  $\approx 6\times$  the frozen control’s rate. **Only one of them is still PING** — the frozen control’s line is bright yellow because pingness stays high ( $\approx 0.9$ ) the whole way; every trainable line is dark purple because pingness was already pruned to  $\approx 0.1$  by the first logged epoch. Reading the colour bar: most of pingness 0.2–0.85 is unused space, because no trajectory crosses it slowly enough to be logged. Reading the geometry: this is the manuscript’s accuracy–rate frontier (nb025, §2.3) animated — and the architecture is the only thing keeping a trained network on the sparse, rhythmic side of it. The dashed divider at  $\approx 17$  Hz marks the two basins explicitly: frozen PING holds the left, every trainable run ends in the right, at the same accuracy.